BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Autumn 1 AP1 Assessment and Review

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before turning to the answers.

1. What does input validation check?

2. How can a larger cache help a CPU?

A) More physical CPU cores
B) Fewer slow memory accesses
C) All RAM becomes non-volatile
D) No instructions need fetching

3. Why is clock speed alone insufficient to compare two CPUs?

4. A quiz needs scores, questions, input and a menu. "Decomposition hides all parts in one unplanned step." Correct this. How should the quiz be split up?

Today's objectives

Recall key ideas from Paper 1 and introductory Paper 2. Show careful working in calculations and traces.

Key vocabulary - Read, cover, recall

register
Small, fast CPU storage holding data or addresses.

compression
Reducing bits needed to represent a file.

abstraction
Ignoring irrelevant problem details.

validation
Checking data against specified input rules.

Write one definition in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Four key ideas

PAPER 1	Recall CPU, storage, binary, images and sound.
PAPER 2	Recall ADA, flowcharts, searching, sorting and validation.
SHOW WORKING	Write formula, values, bits, then bytes.
REVIEW & IMPROVE	Find one strength and one topic to revisit.

SHOW EACH STEP - IMAGE SIZE CALCULATION

PIXELS	BITS	BYTES
50 x 40 = 2000	2000 x 8 = 16000	16000 / 8 = 2000

For calculations: formula -> working -> unit.

I Do - Worked example

Model a synoptic answer: calculate bitmap storage and explain a suitable device for a portable laptop.

1. 50 x 40 = 2,000 pixels.
2. 2,000 x 8 = 16,000 raw bits.
3. 16,000 / 8 = 2,000 bytes.
4. Choose an SSD for a knocked laptop: no moving heads.
5. Drawback: SSDs often cost more per GB than HDDs.

We Do - Three guided questions

1. A 5 x 4 bitmap at 8 bits per pixel uses how many bytes?

2. Linear search for 6 in [2, 5, 6, 1]: give the zero-based index.

3. State what the PC holds and why the FDE cycle repeats.

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the raw number of bits in a bitmap.

```
width = 10
height = 5
depth = 8
print(width * height * depth)
```

1. Predict the output.

2. Change depth to 4.

3. Explain how to convert the result to bytes.

You Do - Independent exam questions

Q2. A 100 x 20 image uses 4 bits per pixel. Calculate its raw size in bytes. [3 marks]

Q3. Explain abstraction and give an example of a detail that could be ignored in a storage-capacity calculator. [3 marks]

Q4. A pupil enters a mark of 105 into a program accepting 0-100. Explain one validation rule and one suitable boundary test. [4 marks]

Exit Ticket - True or False?

Answer before checking Page 4.

1. An unsigned 8-bit binary number can represent denary 300. ☐ T ☐ F

2. To turn raw bits into bytes, divide by eight. ☐ T ☐ F

3. An SSD has moving magnetic read/write heads. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Mark fairly. Tick correct points and fix any mistakes.

Use the mark allocations; then rewrite one answer in full.

Do Now - Answers

Q1 Whether input meets specified rules; it does not prove factual truth or identity. ☐

Q2 B - Fewer slow memory accesses ☐

Q3 Core count, cache and workload also affect performance; comparable evidence is needed. ☐

Q4 Split a large problem into smaller parts. Keep the parts needed for the task, then link clear steps to solve it. ☐

Code First - Model

1. 10 x 5 x 8 = 400 bits. ☐

2. With depth 4: 200 bits. ☐

3. 200 / 8 = 25 bytes. ☐

We Do - Three model answers

1. 5 x 4 x 8 = 160 bits = 20 bytes.

2. Index 2.

3. PC holds next instruction address; repeated cycles process successive instructions.

Exit Ticket - Check

1. FALSE 2. TRUE 3. FALSE

Explain one false statement using the key ideas.

You Do - Self-mark

Q2 100 x 20 = 2000 pixels [1]; x 4 = 8000 bits [1]; / 8 = 1000 bytes [1]. ☐

Q3 Ignore irrelevant detail [1]; retain important detail [1]; device colour need not affect storage [1]. ☐

Q4 Validate mark between 0 and 100 inclusive [2]; test 0 or 100 [1] and accept it [1]. ☐

Try the question again

1. Copy the FULL question you got wrong (or found hardest):

2. In GREEN PEN, write a NEW answer. Include working if needed: